The Representation Shift of World Models: From Explicit Symbolic Maps to Latent Neural Dynamics and LLM-Based Agents

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Driven by advancements in large language models, autonomous agents have gained widespread attention. However, their lack of physical grounding exposes critical structural limitations. Although internal world models are essential for achieving genuine agency, current literature often categorizes these systems based on empirical applications rather than their underlying cognitive architectures. To address this gap, we present a comprehensive review focused on the architectural evolution of world models. We propose a unified conceptual framework to trace the paradigm shift in how architectures represent environmental dynamics, spanning three stages: from human-engineered symbolic representations, to experience-driven latent neural dynamics, and finally to data-driven semantic approximations. This review analyzes the structural evolution across these paradigms to assess their respective strengths and limitations. Furthermore, we outline key challenges and future research directions toward robust, physically grounded artificial intelligence.

Keywords: World Models, Autonomous Agents, Cognitive Architectures, State Representation, Large Language Models

I. Introduction

THE concept of an internal world model has been central to artificial intelligence and cognitive science since Kenneth Craik first proposed that intelligent systems build internal representations of external reality [1]. This idea underpins the development of autonomous agents that use such models to simulate events and plan actions. For decades, the pursuit of designing autonomous agents has been fundamentally connected to the quest to build these internal simulators. Recently, the world model concept has experienced an explosive attention driven primarily by the advent of multimodal large language models such as GPT-4 [36] and Gemini [37]. This resurgence has sparked a debate on whether modern language-based agents truly construct causal internal world models or merely function as sophisticated statistical pattern-matching systems driven by semantic co-occurrence.

However, current literature largely lacks the representational precision required to evaluate the evolution of these agents. While earlier surveys map the internal topologies of agents by categorizing them into classic functional architectures such as model-based, goal-based, or learning agents, they typically treat these internal components as abstract functional blocks [22]. Rather than dissecting the computational medium of the internal state, they focus on the macro-level information flow among sensors, cognitive modules, and actuators. Although structural classifications and domain-specific applications are extensively cataloged, the evolution of the underlying representational mechanisms remains largely unexplored.

Even the most recent surveys fall into this application-centric trap. They attempt to categorize world models by functional utility, distinguishing between understanding of internal state and the prediction of future dynamics[23]. However, they ultimately organize their findings around empirical domains such as autonomous driving, robotics, and social simulation. This functional framework suffers from a critical structural flaw. It conflates fundamentally distinct representational mediums. Consequently, the current literature provides limited clarity on what constitutes a true world model from an architectural perspective.

Existing research generally treats the world model solely as a predictive tool [18] and rarely uses it to analyze the architectural evolution of autonomous agents. To address this gap, we argue that agent evolution is fundamentally driven by a three-stage shift in internal representational mediums rather than purely by increases in computational power. We systematically trace this historical trajectory from human-engineered explicit symbolic maps through experience-driven latent neural dynamics to data-driven semantic approximations.

By charting this historical trajectory, this review makes three primary contributions:

- We propose a unified conceptual framework for architectural evolution. It categorizes world models by representational mediums and state definitions. This formally maps the transition from human-crafted logic to machine emergent systems.
- We trace the architectural transition from symbolic artificial intelligence to deep reinforcement learning. We contextualize the shift toward deep generative world models as a necessary response to environmental noise and high-dimensional data.
- We explore the limitations of current LLM architectures. We contrast statistical pattern matching with genuine internal world models. Since pure token prediction struggles with physical and causal consistency, we outline architectural principles for future trustworthy agents. This provides a blueprint for transitioning from statistical approximations to grounded, structured, and causally consistent intelligence.

The remainder of this review is organized as follows. Section II presents the theoretical foundations of agentic world

models. Section III introduces the proposed framework and its representational timeline. Sections IV–VI analyze three paradigms: symbolic, latent, and semantic models. Section VII discusses open challenges and future directions, and Section VIII concludes the review.

II. Background Concepts

A. *Formalizing the Autonomous Agent: Interaction and the World Model*

We formally define an autonomous agent not merely as a reactive entity, but as an independent system that operates without direct human intervention, maintaining control over its own actions and internal state [16], [17]. It interacts with its environment through a continuous perception-action loop [15] as illustrated in Figure 1.

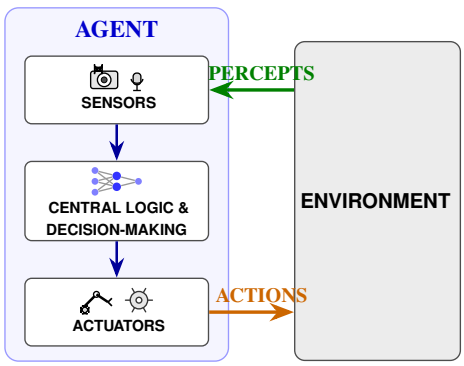

Fig. 1. The continuous perception-action loop of an autonomous agent (Figure adapted from [15]).

In real-world settings, the agent–environment interaction is inherently characterized by partial observability [6]. The agent has no direct access to the true environmental state s_t, and instead receives noisy, high-dimensional observations o_t. This information gap motivates the construction of an internal world model, which acts as a predictive simulator bridging perception and decision-making.

A world model typically serves two key roles:

1) **State Representation:** It encodes the history of observations and actions into a compact latent belief state $\hat{s}_t$, summarizing the agent's understanding of the environment.
2) **Transition Prediction:** It models forward dynamics by predicting future states conditioned on actions:

$$\hat{s}_{t+1} = \mathcal{T}(\hat{s}_t, a_t), \tag{1}$$

where $\mathcal{T}$ denotes the learned transition function.

B. *The Architecture of a World Model: Craik's Pipeline*

To survive in partially observable environments, an agent must actively build an internal simulator [1], [2]. This simulator allows the agent to safely evaluate alternative future actions. Physical mistakes in the real world are often catastrophic and irreversible. Internal simulation provides a safe and highly efficient space for forward planning.

Craik outlined a three-step cognitive process for this predictive capability. It involves translation, inference, and retranslation [1]. This classic framework maps perfectly onto modern computational world models [18], [25], [26]. We can decompose these models into three functional components:

- **Representation Model (Translation)** The agent compresses high-dimensional noisy observations o_t into a compact belief state $\hat{s}_t$ via an encoder function E:

$$\hat{s}_t = E(o_t, h_{t-1}) \tag{2}$$

where h_{t-1} denotes the historical context. This mapping discards irrelevant noise but preserves the essential causal features of the true state s.
- **Transition Dynamics (Inference)** This core engine operates entirely within the internal space. It predicts the future state $\hat{s}_{t+1}$ based on the current state $\hat{s}_t$ and an action a_t:

$$\hat{s}_{t+1} = \hat{\mathcal{T}}_\theta(\hat{s}_t, a_t) \tag{3}$$

where $\hat{\mathcal{T}}_\theta$ approximates true environmental dynamics. The agent uses this to unroll future trajectories without physical execution.
- **Output Model (Retranslation)** The model projects internal predictions back into actionable metrics. This step evaluates the utility of the plan. It typically uses a reward predictor $\mathcal{R}_\phi$ and an observation decoder $\mathcal{Z}_\psi$:

$$r_{t+1} = \mathcal{R}_\phi(\hat{s}_{t+1}), \quad \hat{o}_{t+1} = \mathcal{Z}_\psi(\hat{s}_{t+1}) \tag{4}$$

These metrics ground the simulated trajectories in reality. They directly guide the agent to select the optimal action.

C. *The Key: State Representation as the Computational Medium*

The basic architecture of world models remains historically consistent. As illustrated in Figure 2, the encoder, transition engine, and decoder form a stable computational pipeline [38]. However, decades of research diverge on one fundamental variable. That variable is the representation of the internal state $\hat{s}_t$. The true environmental state s is infinitely complex. To handle this computationally, agents must project the reality onto a simplified computational medium. Crucially, this internal state $\hat{s}_t$ is not limited to passively representing the world alone. Instead, it typically encodes a predictive belief state, which includes not only a compressed history of the environment, but also relevant information about future actions and their potential consequences.

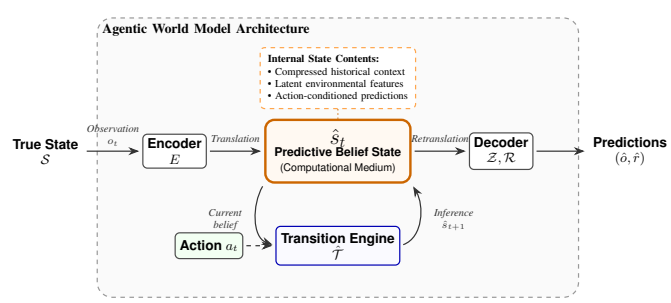

Fig. 2. **Architecture of the Agentic World Model Pipeline.** The internal state $\hat{s}_t$ functions as the central *computational medium*. It intrinsically encodes compressed history, latent physical features, and predictive evaluations of future action consequences.

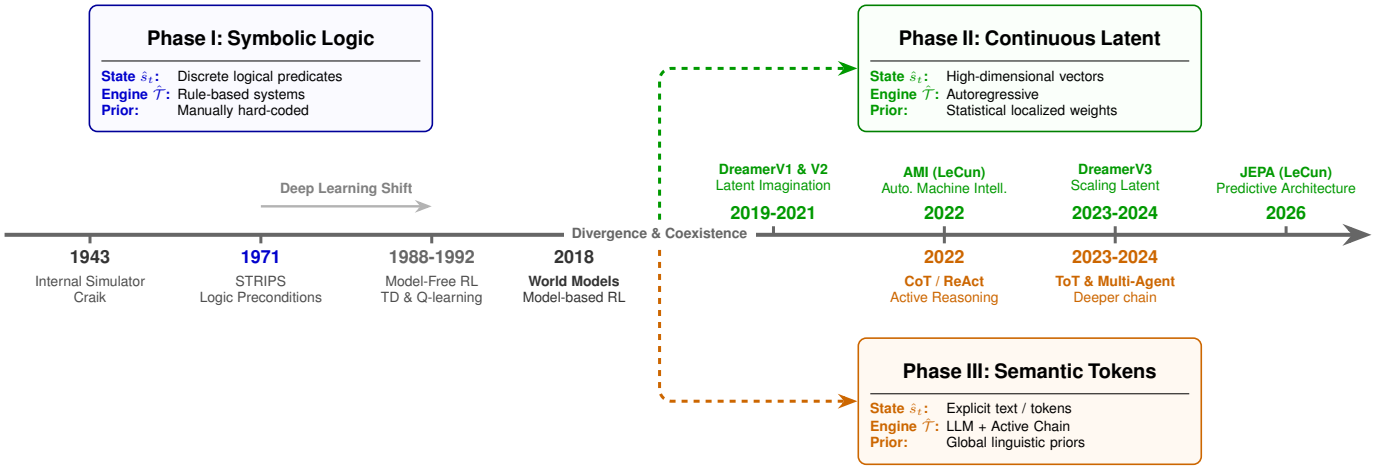

Fig. 3. **Evolution and Coexistence of Agentic World Models.** A detailed timeline demonstrating the architectural paradigm shifts from 1943 to the present. After the formalization of World Models in 2018, the trajectory bifurcates. Phase II (Continuous Latent dynamics, championed by Dreamer and LeCun's AMI/JEPA) and Phase III (Semantic Tokens, driven by LLMs) are not linear replacements but rather parallel, coexisting paradigms competing to build physically grounded intelligence.

III. Summary of the Representational Evolution

As shown in the timeline in Figure 3, the architecture of autonomous agents has undergone significant evolution. Over the past half-century, the field progressed from the rigid, deterministic symbolic systems of the 1970s in Phase I to the robust neural simulations established around 2018 in Phase II. Crucially, this trajectory does not follow a strict linear substitution. The latent dynamics of Phase II co-evolve and coexist alongside Phase III, a contemporary paradigm driven by the semantically rich, discrete token spaces of Large Language Models.

IV. Phase 1: Explicit Symbolic Maps

The foundational period of artificial intelligence constructed world models as completely closed, explicit symbolic systems [5]. Rather than dynamically reflecting physical reality, these early architectures functioned as static, explicitly predefined logical mazes. By permanently decoupling the agent from real-time environmental feedback, their interaction became purely computational. The agent did not organically adapt to a changing world, but instead executed search algorithms to compute predefined paths as shown in Figure 4. Trapped within these frozen boundaries, the system lacked any mechanism to update its internal representations against unpredicted physical events.

A. *State & Transition*

The internal state representation s_t consisted strictly of human-engineered symbolic representations. The computational medium was a discrete collection of logical predicates and Boolean variables [3], [4]. Agents did not learn these representations from sensor data. Human programmers manually defined the state space. For example, a physical arrangement of blocks had to be hard-coded into strict spatial relations. Figure 5 visualizes this architecture, illustrating how the environmental observations were grounded into discrete predicates before entering the dynamic engine. This discrete

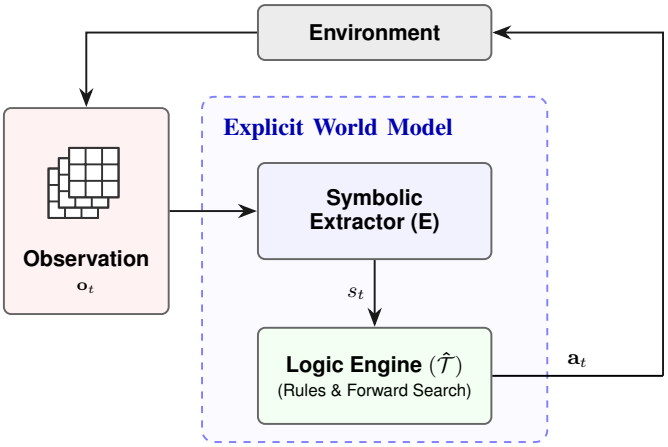

Fig. 4. Flow diagram of the Phase I explicit symbolic architecture.

medium provided perfect causal interpretability but required extensive manual engineering for each new environment.

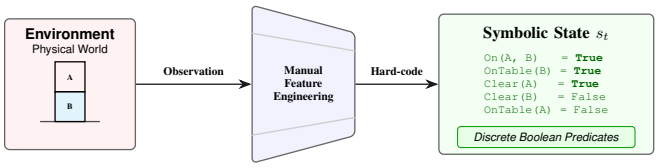

Fig. 5. Phase I Symbolic State Representation.

The transition model T was disconnected from real-time feedback and functioned as a deterministic forward search engine. Given a state $\hat{s}_t$ and action a_t, it ignored probabilities and applied fixed preconditions with simple add and delete rules to produce $\hat{s}_{t+1}$ [3] shown in Figure 6.

B. *Structural Limitations of the Explicit Symbol Paradigm*

However, this explicit architecture performed poorly in the physical world. The first limitation was the symbol grounding problem [7], [8]. Real sensors produce continuous, noisy

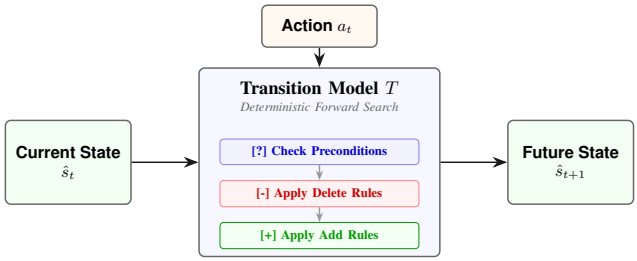

Fig. 6. Phase I Symbolic Transition Model.

data. The perception module cannot compress this noise into discrete Boolean symbols. The logic engine demanded absolute certainty. Even small perceptual errors could propagate through the pipeline, leading to unreliable planning outcomes.

Second, the architecture suffered from computational intractability [9]. The system lacked scalability. Complex environments triggered exponential growth in the logical search tree. This combinatorial explosion quickly exhausted the hardware limits and the search process.

Third, the frame problem crippled the paradigm [10]. All rules had to be manually specified for each possible physical interaction. Moreover, they had to explicitly define which aspects of the world remained unchanged after each action. Such exhaustive manual engineering is impractical in real-world settings.

V. Phase 2 Experience Driven Latent Neural Dynamics

Although explicit symbolic representations have limitations, they can still be effective in closed environments. However, in open environments, which are inherently uncertain, symbolic approaches are not well suited. Under uncertainty, probabilistic representations are introduced, allowing different states of the world to be assigned quantitative values. At this stage, there are differences in the process of interaction between agents and the environment to decision-making.

Early research on agent–environment interaction focused on model-free reinforcement learning [11], [12]. These algorithms map inputs directly to actions bypassing the prediction of the future state. Agents learn through direct interaction with the physical world. However, such trial-and-error processes are often dangerous and costly. For example, incorrect actions by a robot may damage hardware. In addition, these methods suffer from poor sample efficiency. Subsequently, research shifted toward model-based reinforcement learning [14], [18]. These approaches learn a dynamic model of the environment to predict future state transitions and generate "imagined" trajectories for planning and decision-making. Compared to direct interaction, they reduce sampling costs and mitigate risks associated with real-world trial and error. However, they introduce the issue of error accumulation, particularly in long-term predictions [25]. Such errors may compound over time, ultimately degrading decision quality.

David Ha and Jürgen Schmidhuber introduced the World Models framework in 2018, laying the foundation for learning latent dynamics. This framework is inspired by human cognition, where individuals construct internal models of the world based on limited sensory input, forming abstract spatiotemporal representations [18]. As illustrated in Figure 7, the framework consists of three core components: a Vision model (**V**), a Memory model (**M**), and a reinforcement learning Controller (**C**). By shifting trial-and-error processes into a learned latent space, the agent can simulate future outcomes based on current actions, making learning significantly safer and more sample-efficient than direct physical interaction.

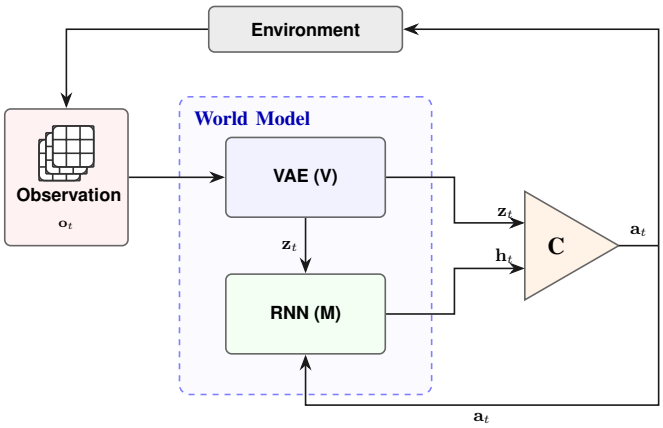

Fig. 7. Flow diagram of the Phase II latent space agent architecture.

A. *State & Transition*

At this stage, the representation shifted away from discrete symbolic predicates. Early approaches relied on manually designed features [13], such as spatial coordinates. However, manual feature engineering is difficult to scale and requires substantial effort. When dealing with high-dimensional raw pixel inputs from RGB cameras, these approaches also suffer from the curse of dimensionality. As illustrated in Figure 8, the introduction of deep neural encoders enabled the use of models such as the Variational Autoencoder (VAE) [19]. These models compress high-dimensional, multi-channel inputs into dense, continuous latent representations. Such representations are well suited for capturing stochasticity in the physical world while significantly reducing reliance on manual annotation. However, this shift comes at the cost of interpretability, as the learned latent space is continuous but difficult to interpret semantically.

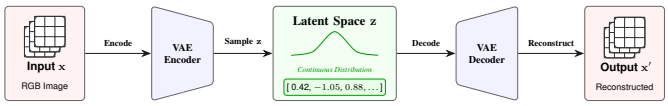

Fig. 8. Phase II Latent Space State Representation.

The system learns a state transition model to characterize the temporal evolution of the environment. In latent space, the dynamics are typically modeled as:

$$z_{t+1} = f(z_t, a_t) \tag{5}$$

Early world model approaches commonly employ recurrent neural networks (RNNs) [24] to capture sequential dependencies. A representative example is the latent dynamics model

proposed by David Ha and Jürgen Schmidhuber [18]. As illustrated in Figure 9, such architectures capture temporal dependencies and enable multi-step trajectory rollouts entirely in latent space.

Through autoregressive prediction, the model can simulate future trajectories and evaluate the long-term effects of action sequences without interacting with the real environment. This internal simulation mechanism improves safety and enhances sample efficiency.

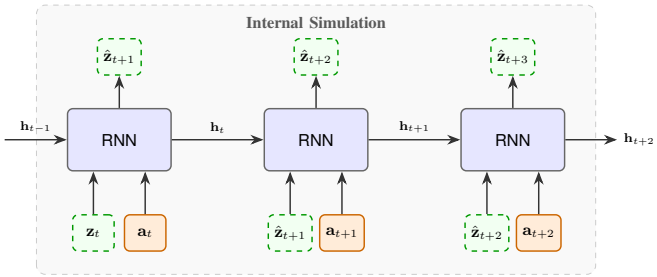

Fig. 9. Phase II Latent Space Transition Representation.

B. Frontier Directions

Following the initial formalization of latent world models, the field has expanded significantly. One important development can be traced back to the world model framework proposed by David Ha and Jürgen Schmidhuber [18], which introduced the idea of learning compact latent dynamics for control. Building on this paradigm, the Dreamer series has progressively addressed the limitations of early continuous latent state models by improving both representation learning and world model design, thereby advancing generative latent control.

Building upon the Recurrent State Space Model (RSSM), DreamerV1 [25] demonstrated that agents can learn effective policies entirely within a learned latent imagination. However, continuous latent representations often struggle to capture sharp, discrete transitions, leading to error accumulation over long horizons. To address this, DreamerV2 [26] introduced discrete categorical latents, replacing continuous Gaussian states. This discrete representation acts as a bottleneck that filters high-frequency noise and improves performance in environments with discrete structure, such as Atari[1].

Despite this advancement, DreamerV2 remains sensitive to variations in reward and observation scales, requiring careful hyperparameter tuning. DreamerV3 [27] mitigates this issue through a scale-invariant design, including symlog transformations and normalized prediction targets. By stabilizing representation scales, DreamerV3 achieves more robust and domain-agnostic performance, successfully scaling to diverse environments such as Minecraft without task-specific tuning.

In parallel to this generative, control-oriented paradigm, a non-generative trajectory has also emerged. This line of work is rooted in Yann LeCun's Autonomous Machine Intelligence (AMI) framework [38], first proposed around 2022, and further developed through Joint-Embedding Predictive Architectures (JEPA), such as LeWorldModel [39] in 2026. This perspective argues that reconstructing high-fidelity observations may waste modeling capacity on unpredictable or irrelevant details. Instead, JEPA-based approaches predict future states directly in a shared latent space. Recent work such as LeWorldModel further shows that such representations can capture meaningful physical structure, enabling the model to encode and probe physical properties of the environment. Although this direction is less mature than reinforcement learning-based generative models, it provides a complementary perspective on latent world models.

[1]Atari refers to a suite of classic arcade games commonly used as benchmarks in reinforcement learning, where agents learn to play directly from pixel inputs.

C. Structural Limitations of the Latent Space Paradigm

Despite the impressive advancements in both generative and non-generative trajectories, the broader Phase II paradigm continues to encounter fundamental structural limitations. The first critical limitation is the loss of causal interpretability [20]. While discrete symbolic models were perfectly transparent, the continuous high-dimensional vectors in Phase II function essentially as black boxes. Although these vectors effectively manage perceptual noise, they lack explicit semantic meaning, making it challenging for humans verify the internal logic of the agent's decisions.

Second, the latent paradigm still faces a severe generalization limitations regarding global common sense [21]. Although recent frontier models have improved physical and task adaptability, DreamerV3 handles vastly different tasks (from Atari to Minecraft) with high robustness. It achieves this through scale-invariant designs, such as symlog transformations and two-hot encoding. These specific techniques effectively stabilize extreme variations in rewards and observations. Similarly, the non-generative LeWorldModel [39] significantly improves physical perception. It successfully trains a JPEA architecture directly from raw pixels. As a result, it can encode meaningful physical structures and detect physically implausible events. This breakthrough successfully narrows the physical grounding gap.

However, physical and scale generalization do not equal semantic common sense. The internal priors of latent agents rely strictly on localized statistical weights. They learn purely through direct environmental trial-and-error. Therefore, they operate completely without the pre-trained, broad worldly knowledge of humans. Transfer of these latent policies to novel environments with different semantic rules remains highly restricted. To address this limitation, researchers have begun integrating linguistic priors, which has directly motivated the transition toward the discrete semantic token paradigm in Phase III.

VI. Phase 3: Data-Driven Discrete Semantic Approximations

The emergence of Large Language Models brought renewed attention to the classical concept of autonomous agents. Instead of relying on uninterpretable, task-specific latent engines, researchers adopted LLMs as central cognitive brains. Human

language naturally encapsulates broad common sense, logical priors, and worldly dynamics. Consequently, the primary representation medium shifted from continuous vectors to explicit multimodal tokens. As illustrated in Figure 10, this shift fundamentally redefined the agentic world model. A perception module translates diverse inputs into a unified semantic sequence, which is maintained within a dynamic context window. Rather than reacting reflexively, the LLM leverages this historical prompt to perform explicit external planning. It utilizes Chain-of-Thought (CoT) reasoning to evaluate intermediate steps before generating an optimal action. This action is then simultaneously executed in the environment and appended back into the memory loop, creating a fully interpretable, self-evolving cognitive cycle.

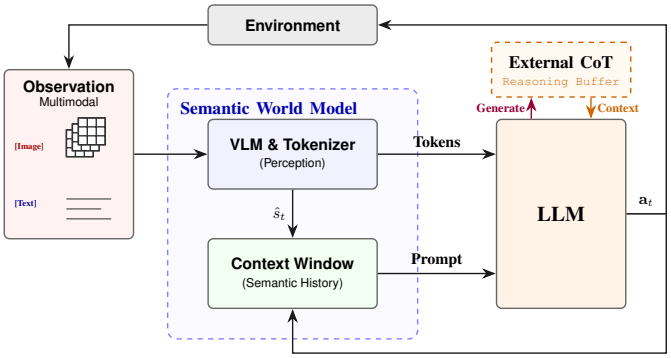

Fig. 10. Flow diagram of the Phase III LLM-based agent architecture.

A. *State & Transition*

Following this architectural shift, the state representation $\hat{s}_t$ underwent a fundamental transformation. As shown in Figure 11, the system abandoned the implicit, uninterpretable hidden weights of Phase II [18], [25], [26]. Instead, it transitioned to explicit multimodal sequences. Modern Vision-Language Models (VLMs) process distinct inputs independently. Vision encoders patch raw visual pixels, while tokenizers encode text instructions. Finally, they are flattened into a single interleaved sequence of discrete tokens [35]. This explicit multimodal context window significantly improves interpretability. Because the working memory now consists of readable linguistic and visual markers, humans can directly inspect and modify the agent's internal state.

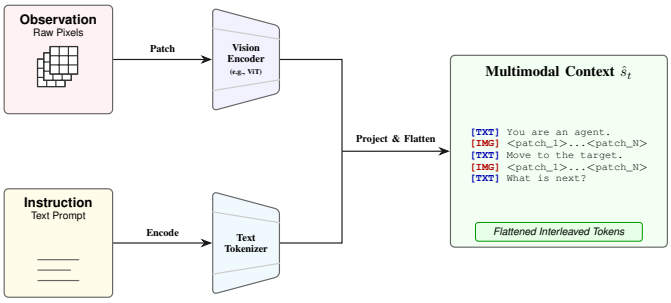

Fig. 11. Phase III Multimodal State Representation.

To process these semantic states, the transition engine $\hat{\mathcal{T}}$ evolved through a distinct structural lineage. Initially, systems utilized standard conversational prompting. The LLM functioned as a direct input-output mapper, passively generating actions from text observations. To overcome this simple reflex, researchers introduced Chain-of-Thought (CoT) [31]. CoT compelled the agent to articulate multi-step internal reasoning before physical execution, shown in Figure 12. The transition function became an active process of deductive generation. However, pure CoT remained isolated from environmental feedback. This limitation motivated the development of ReAct framework [28]. ReAct formally interleaved internal reasoning with external acting. It allowed the central brain to iteratively test hypotheses and update its context based on real-world observations. From this foundational loop, the architecture expanded into two advanced directions. For complex planning, Tree of Thoughts (ToT) [29] enabled the engine to systematically search and backtrack across multiple reasoning branches. For comprehensive problem-solving, Multi-Agent collaborations [30] distributed the cognitive load among specialized LLM roles. These external mechanisms successfully transformed a static text generator into an interactive, dynamic transition engine.

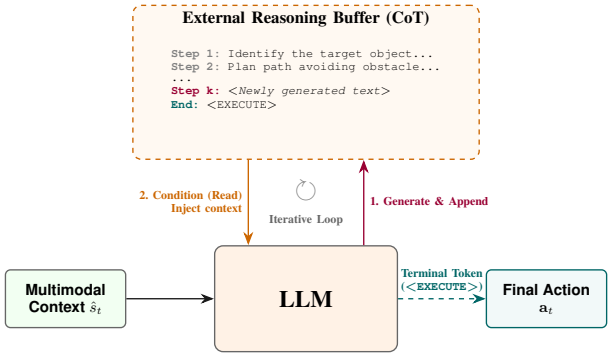

Fig. 12. Phase III LLM-based Transition Representation.

B. *Structural Limitations of the LLM-based Paradigm*

Despite enabling cross-domain generalization, the LLM-centric paradigm introduced its own structural limitations. Fundamentally, a purely semantic model fails to acquire genuine abstract representations of space and time. Language models excel at high-level logic, but lack an intrinsic understanding of continuous geometry [34]. An agent can logically plan to stack irregular objects but fail to account for spatial constraints such as friction or the center of mass.

Second, the architecture struggles with temporal and causal consistency. Because the engine predicts statistically likely tokens rather than simulating true physical dynamics, it is highly prone to hallucinations [32]. The generated transitions often fail to align with the real-world dynamics.

Third, the system is constrained by a bounded context capacity. As environmental interactions accumulate, the prompt length expands aggressively. This inevitably degrades long-context attention [33]. Unlike the recursive memory of continuous latent engines, this rigid text window severely limits the agent's ability to maintain coherent long-horizon autonomy.

VII. Open Problems and Future Directions

While Phase III has equipped autonomous agents with unprecedented semantic reasoning capabilities through Large Language models, the transition from simulated environments to open physical worlds still presents significant unresolved challenges. The evolution of the agentic world model is far from complete. We identify several critical open problems and future directions.

A. *The Illusion of Understanding: Autoregressive Pattern Matching vs True World Models*

Fundamentally, all models are analogical abstractions. By construction, they simplify reality and inevitably fail to capture certain aspects of the underlying system [1]. In the era of large language models, agents are often built by treating these models as surrogate "cognitive processors" that receive environmental input and generate decisions. While LLMs are powerful tools, this paradigm introduces a critical illusion of understanding.

The effectiveness of a world model depends entirely on whether its internal relational structure correctly parallels external physical processes [1]. However, modern LLMs operate primarily through statistical next-token prediction, subsequently refined by human preference alignment techniques. While these alignment processes make the generated text exceptionally coherent, persuasive, and conversational, they optimize for human approval rather than objective physical truth. Unlike human cognition, LLMs lack explicitly grounded common sense or structural mechanisms for causal reasoning [38]. Consequently, their outputs frequently exhibit an "illusion of understanding"—appearing highly plausible to human evaluators while fundamentally lacking true alignment with underlying environmental dynamics.

Crucially, autonomous agents must predict the spatiotemporal evolution of the physical world, including the dynamic consequences of their physical actions, rather than merely predicting the most statistically favored semantic token. This profound mismatch between conversational alignment and physical state-level reasoning lies at the heart of the hallucinations and physical inconsistencies observed in current LLM-based agents.

Future Direction: Future research may benefit from revisiting the ideas of LeCun and Craik, who emphasize structured world models and predictive representations [1], [38]. Historically, major technological advances have extended human sensory, cognitive, or physical capabilities. Following this trajectory, building AI systems inspired by the structure and function of the brain may offer a promising path forward.

At the macro level, this suggests viewing intelligence through a modular perspective. The brain can be understood as a coordinated system of interacting components, each responsible for distinct functions such as perception, memory, reasoning, and action. Future agent architectures may benefit from explicitly modeling modular organization. They should also capture how different modules communicate and coordinate to produce coherent behavior.

At the micro level, deeper investigation is required into how each module operates internally. First, although Large Language Models exhibit clear limitations, they remain a powerful extension of human language capabilities. The language itself serves as an effective abstraction to represent complex concepts. In future architectures, language is likely to remain an important medium for information processing, but not the only one. This points toward the need to explore richer and more expressive representations beyond language, enabling models to better capture physical, spatial, and causal structure.

Second, insights from Daniel Kahneman's dual-system theory offer a useful framework for reasoning [40]. System 1 corresponds to fast, intuitive, pattern-based processing, which closely resembles the behavior of current LLMs. While System 2 involves slow, deliberate, and structured reasoning. Future agent designs may benefit from explicitly incorporating both modes. A fast, learned component for efficient pattern recognition, and a slower, more structured component for planning, verification, and logical inference. The integration of these two systems could enable more robust, reliable, and interpretable decision-making.

B. *Computational limitations in Long-Horizon Planning*

A fundamental challenge in long-horizon planning arises across both experience-driven latent world models and LLM-based agents, but for different reasons. In latent dynamics models built on recurrent architectures, memory is typically maintained through RNN hidden states [24], [25]. Although efficient in principle, such mechanisms suffer from well-known limitations in long-term dependency modeling, including information decay, state aliasing, and difficulty in preserving precise historical structure over extended horizons. As a result, the learned latent state may gradually lose fidelity, leading to compounding prediction errors over time.

In contrast, LLM-based agents rely on explicit context windows to store past interactions. This introduces a different limitation: the inability to scale beyond a fixed memory capacity [33]. As task horizons grow, histories of states, actions, and observations must be either truncated or externally managed. In addition, planning strategies such as Chain-of-Thought [31] or Tree of Thoughts [29] introduce significant computational overhead, making them impractical for real-time decision-making.

Future Direction: A proposed direction is memory abstraction and compression, which aims to bridge these limitations. Rather than storing raw trajectories, agents should learn structured, hierarchical representations of experience. This involves: (1) extracting high-level latent summaries that capture task-relevant semantics while discarding low-level noise, (2) organizing memory into multiple timescales (short-term and long-term memories), and (3) maintaining compact representations of key spatial and causal relationships that remain stable over time.

Such a framework would enable agents to selectively retain task-relevant information while discarding redundant details. By combining efficient summarization with hierarchical memory organization, it becomes possible to significantly reduce

computational burden without sacrificing decision quality. It provides a viable pathway toward scalable and real-time long-horizon planning.

VIII. Conclusion

In this review, we propose a unified framework to trace the evolution of agentic world models through three stages: explicit symbolic systems, experience-driven latent dynamics, and LLM-based semantic models. Viewed from the perspective of world models for decision-making agents, these paradigms can be understood as progressively richer mechanisms for representing the environment and supporting future prediction. Each stage reflects a different trade-off in how an agent constructs internal models of the world to support perception, reasoning, and planning. We argue that none of these paradigms is sufficient on its own for building robust autonomous agents in complex environments. Future progress should therefore focus on integrating their complementary strengths within a unified agent framework. In particular, effective world models for autonomous agents must jointly support causal structure, physical grounding, and semantic reasoning, enabling a transition from purely statistical prediction toward more reliable and grounded decision-making and future prediction systems.

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